V1.3 (2020)

正式售卖的第一个版本

V2.0

内部迭代，未销售

V2.1 (2021)

电源开关更换为软开关，避免电弧造成开关损耗
稳压方案更换为SY8205
增加2个舵机接口，与电机驱动信号接口2复用
增加USART4接口，方便连接OpenART mini
HC-SR1一体式超声波接口变更为3.3V供电
增加CR引脚延时泄放二极管

V2.2 (内部迭代)

减小电机PWM信号上串接的电阻为100R
稳压方案替换为SCT2450
用户按键及拨码开关增加100R串联电阻
USART4 USART1 无线串口接口，增加供电时序控制电路
软开关更改为P-MOS，断VCC的方案

V2.3 (2022)

更正软开关电路
USART1 USART4 丝印标注更改为 UART1 UART4
UART1 UART4 无线串口接口的5V引脚丝印标注更改为5V*

V3.0 (2022)

5V稳压和舵机稳压更换为SP1N28STER
5V稳压输出端增加SS54二极管
去掉FlexCom摄像头接口
重新分配电机、舵机、编码器、按键、拨码开关引脚。
增加SPI-无线模块接口
增加磁铁检测模块接口

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Title: RT1064_MotherBoard		**Author:** 追猎者	
Subtitle: Revision		**Revision:** V3.0.1.2	
Size: A4	**Date:** 2024/3/25	**Time:** 11:18:16	**Sheet:** 1 **of:** 6

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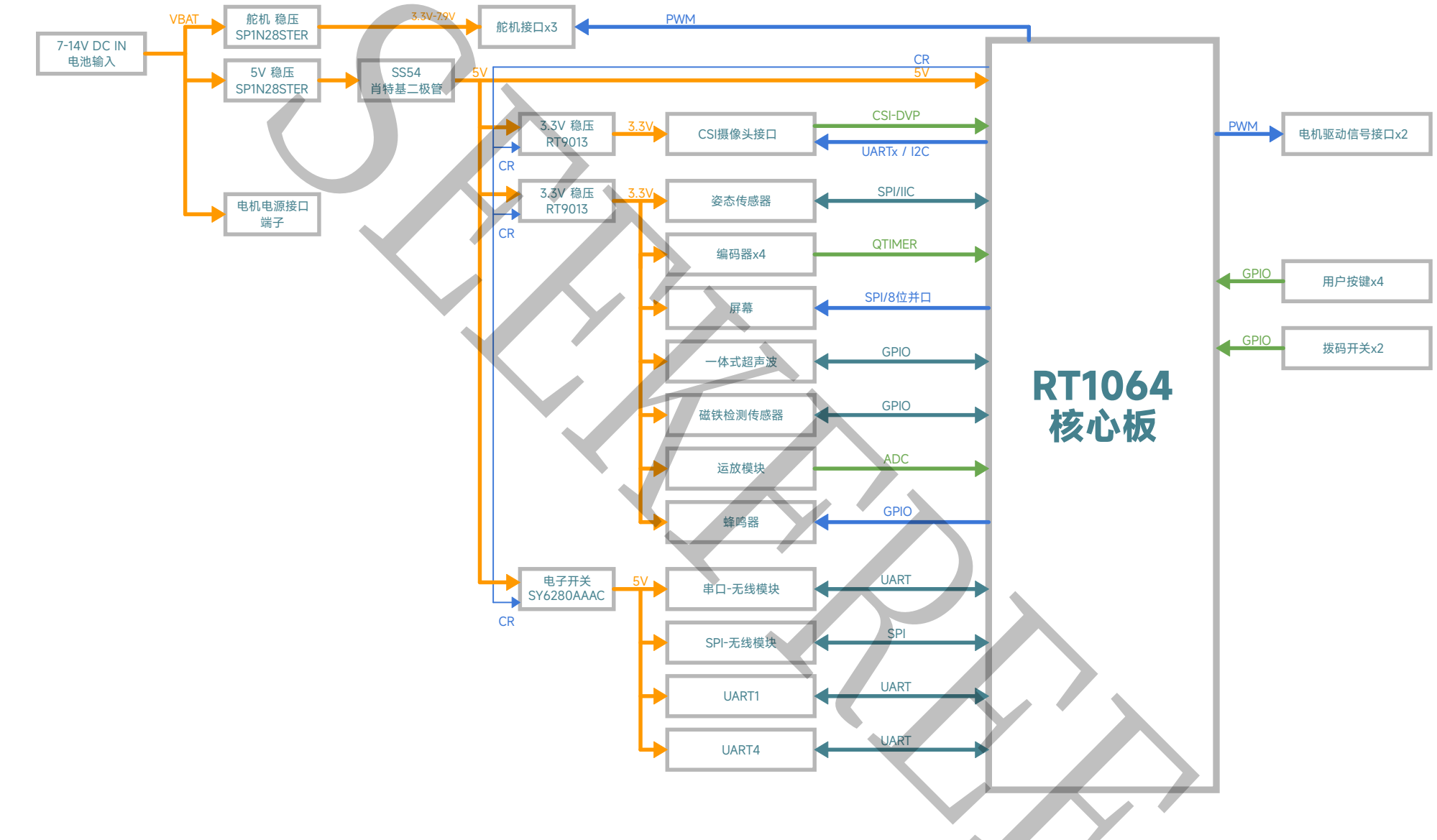

图例:

双向信号: UARTx　输出信号: UARTx　输入信号: UARTx　电源: 5V

红色字体: 需要注意/未确定条目

Title: RT1064_MotherBoard			Author: 追猎者
Subtitle: BlockDiagram			Revision: V3.0.1.2
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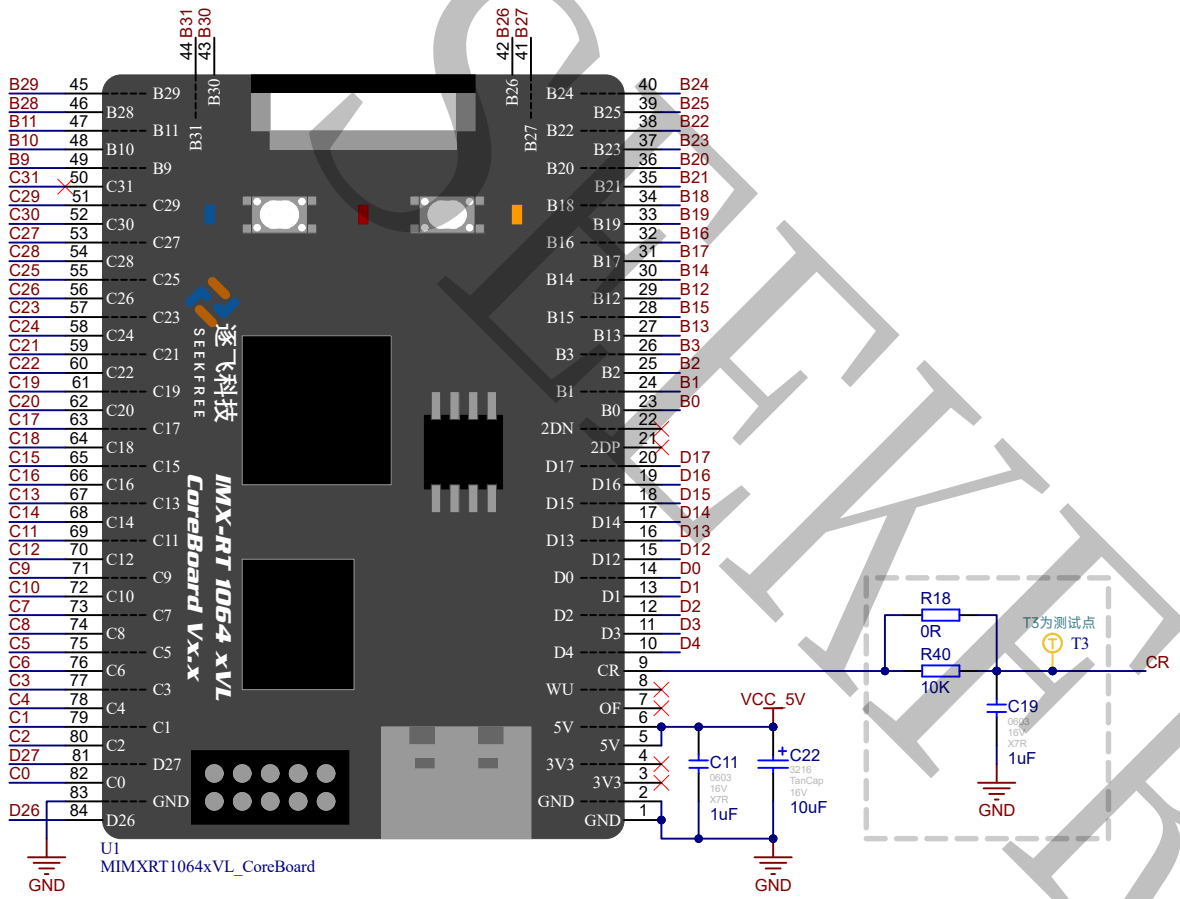
RT1064核心板



引脚表

外设	功能	引脚
CSI 摄像头	PCLK VSY HREF TXD RXD **D0-D7**	B20 B22 NC C28 C29 **B31-B24**
屏幕 接口	SCK MOSI RST DC CS BLK D0-D3 D4-D7	B0 B1 B2 C19 B3 C18 B16-B19 D12-D15
编码器1	LSB DIR	C0 C1
编码器2	LSB DIR	C2 C24
编码器3	LSB DIR	C3 C4
编码器4	LSB DIR	C5 C25
电机1/2	PWM	C6 C7 C8 C9
电机3/4	PWM	C10 C11 D2 D3
舵机1	PWM	C30
舵机2	PWM	D0
舵机3	PWM	D1

外设	功能	引脚
串口 无线 模块	Tx Rx RTS CMD	D16 D17 D26 D27
SPI 无线 模块	SCK CS MOSI MISO RST INT	D12 D13 D14 D15 B16 B17
姿态 传感 模块	SCK MOSI MISO CS	C23 C22 C21 C20
运放 模块	ADC	B14 B15 B21 B23
UART1	TXD RXD	B12 B13
UART4	TXD RXD	C16 C17
拨码 开关	GPIO	C27 C26
按键	GPIO	C15 C14 C13 C12
蜂鸣器	GPIO	B11
磁铁检测	GPIO	D4
一体式 超声波	Trig Echo	B9 B10

TIPS：

1.主板上的所有外设（除舵机外），均使用CR引脚来控制上电时序，

2.若使用RT1064 V1.2版本核心板，需去掉R18电阻，保留R40 C19，以启用外置阻容延时电路。

3.若使用RT1064 V1.3及更高版本、或RT1064 Lite任何版本核心板，
在画自己的主板时，可以省略虚线框里的全部器件。

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Title: RT1064_MotherBoard		Author: 追猎者	逐飞科技 SEEKFREE
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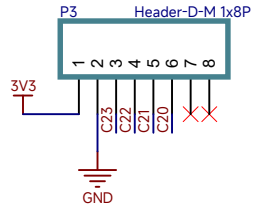
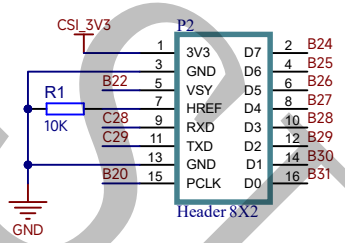
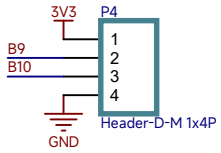
姿态传感模块接口



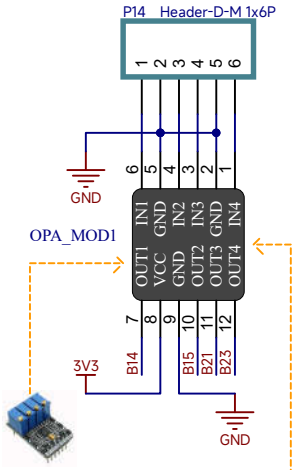
CSI摄像头接口



一体式超声波接口

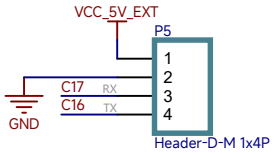


运放模块接口

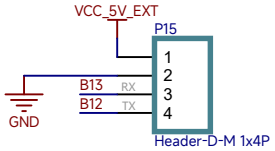


1.OPA_MOD 这个黑色方块是逐飞科技OPA4377运放模块插座，在PCB上对应器件为两个1x6P直插排座。并不是运放芯片！
2.参赛选手自己画原理图时，可以将运放模块电路集成在主板上；也可以自行制作运放模块，只在主板上放模块接口。

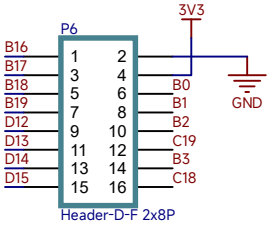
UART4接口



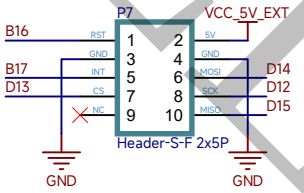
UART1接口



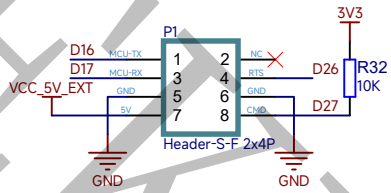
屏幕接口



SPI-无线模块接口

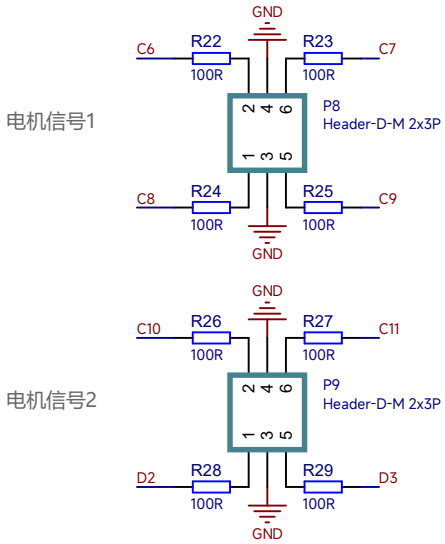


串口-无线模块接口

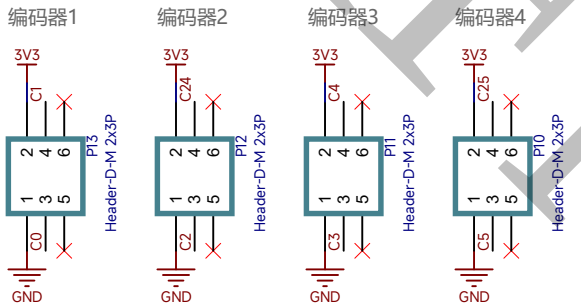


电机驱动接口

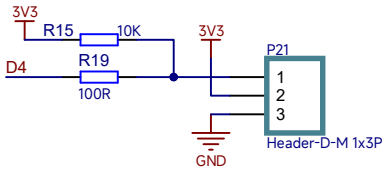
电机接口



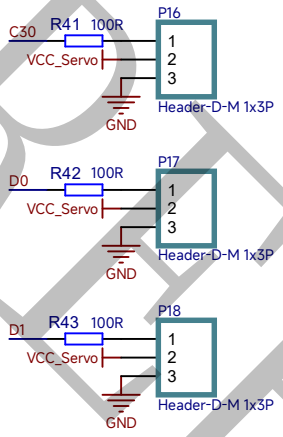
编码器接口



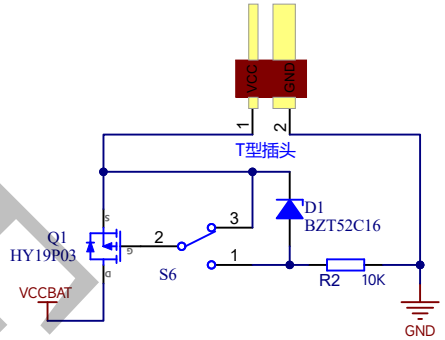
磁铁检测模块接口



舵机接口



电源开关



此部分电路作用为解决开关打火损耗

Title: RT1064_MotherBoard		Author: 追猎者	
Subtitle: Interface		Revision: V3.0.1.2	
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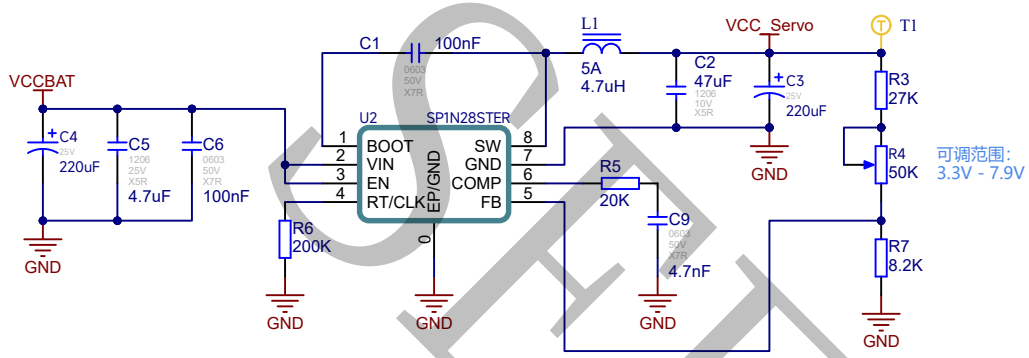
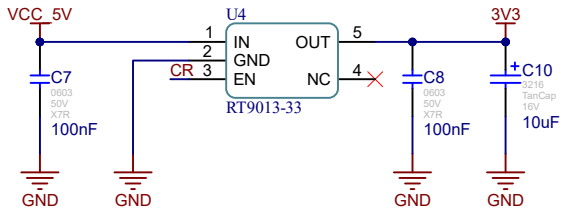
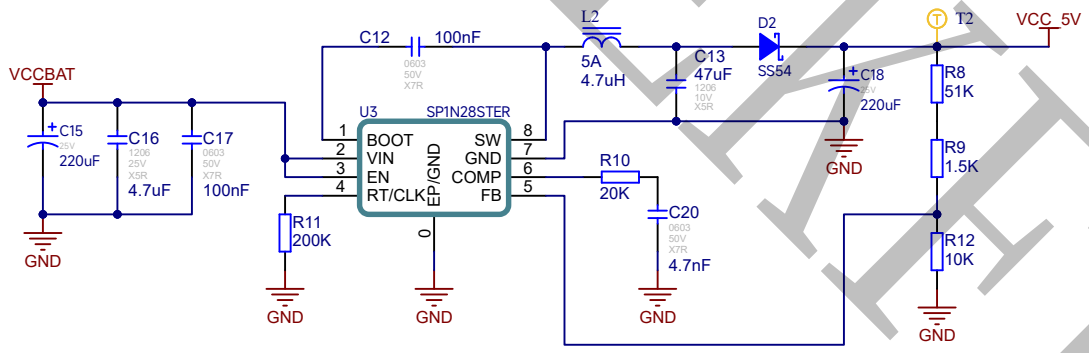
舵机稳压电路



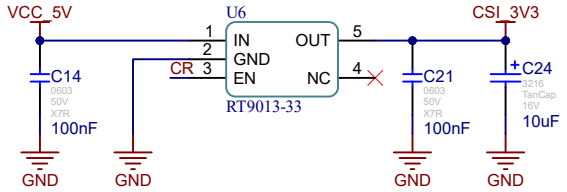
3V3电源



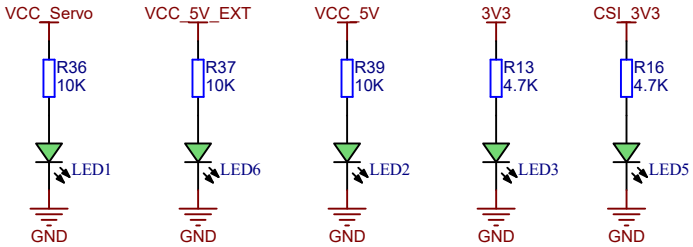
5V稳压电路



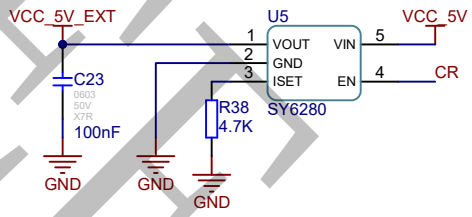
CSI摄像头电源



电源指示灯



5V外设上电时序控制



Title: RT1064_MotherBoard		**Author:** 追猎者	
Subtitle: VRM		**Revision:** V3.0.1.2	
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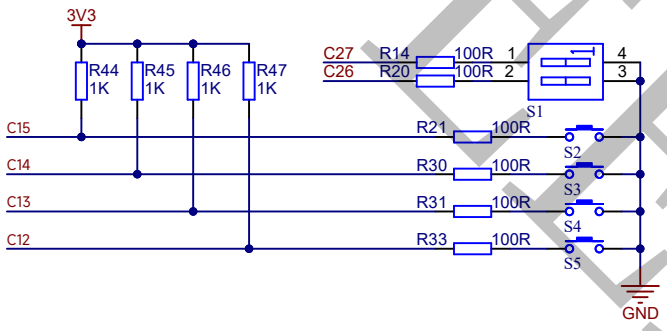
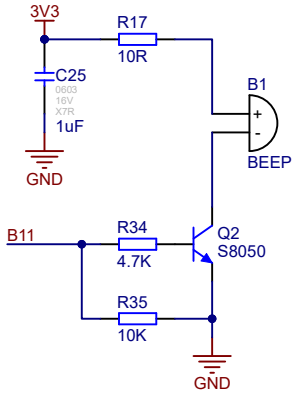
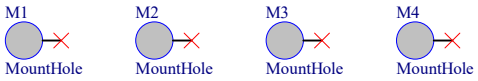
拨码开关/按键



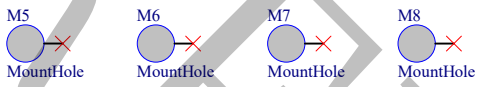
蜂鸣器



固定孔



摄像头底座固定孔



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Subtitle: MISC		**Revision:** V3.0.1.2	
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